ERROR IDENTIFICATION LOG

RiskLens Technologies - Project 1A

Deliberate errors embedded in the Zetheta training material, with the corrected interpretation and supporting source.

Document control	Value
Version	2.1
Errors reviewed	13 / 13
Source basis	Zetheta Project 1A training material + assessor appendix
Entry structure	Incorrect statement why it is wrong correction source RiskLens note

How to read this log: each error is recorded separately so the assessor can check the original claim, the correction, the reasoning and the source.

Quotes are limited to the identifying sentence or claim rather than reproducing the training material in full.

1. Nielsen's heuristics: 8 vs 10 · A1.1

Assessor-facing record of one deliberate error in the Zetheta training material.

INCORRECT STATEMENT

“Jakob Nielsen’s 8 Usability Heuristics, published in 1994.”

WHY IT IS WRONG

The statement misstates both the count and the historical framing. Nielsen's established set contains 10 usability heuristics, including Recognition Rather than Recall.

CORRECT INFORMATION

Nielsen's interface guidance is commonly presented as the 10 Usability Heuristics. The heuristics were published in their well-known form in 1994; Recognition Rather than Recall is heuristic #6.

SOURCE / VERIFICATION

Nielsen Norman Group, “10 Usability Heuristics for User Interface Design” — https://www.nngroup.com/articles/ten-usability-heuristics/

RISKLENS NOTE

The useful design implication is unchanged: make important information visible so users can recognize it instead of having to recall it.

2. Fitts's Law date · A1.2

Assessor-facing record of one deliberate error in the Zetheta training material.

INCORRECT STATEMENT

“Fitts’s Law, formulated by Paul Fitts in 1964…”

WHY IT IS WRONG

The cited year is wrong. The original paper was published in 1954.

CORRECT INFORMATION

Paul M. Fitts published the foundational study The Information Capacity of the Human Motor System in Controlling the Amplitude of Movement in 1954 in the Journal of Experimental Psychology.

SOURCE / VERIFICATION

Fitts, P. M. (1954), Journal of Experimental Psychology, 47(6), 381–391. DOI 10.1037/h0055392 — https://doi.org/10.1037/h0055392

RISKLENS NOTE

For RiskLens, the practical UX point is still relevant: larger, appropriately placed targets reduce movement difficulty.

3. VaR and coherence · A1.3

Assessor-facing record of one deliberate error in the Zetheta training material.

INCORRECT STATEMENT

“VaR … satisfies all four of these properties as well…”

WHY IT IS WRONG

VaR does not, in general, satisfy sub-additivity, so it is not a coherent risk measure in general.

CORRECT INFORMATION

VaR does not generally satisfy sub-additivity. Expected Shortfall / CVaR is used as a coherent alternative under appropriate definitions. This distinction matters when communicating portfolio aggregation and diversification.

SOURCE / VERIFICATION

Acerbi & Tasche, “On the coherence of expected shortfall” (Journal of Banking & Finance, 2002); Inui & Kijima (2005). — https://doi.org/10.1016/S0378-4266(02)00283-2

RISKLENS NOTE

The RiskLens UI should therefore avoid implying that VaR and CVaR have identical coherence properties.

4. Sharpe-ratio annualisation · A1.3

Assessor-facing record of one deliberate error in the Zetheta training material.

INCORRECT STATEMENT

“sqrt(T) applied to the ratio directly.”

WHY IT IS WRONG

Annualisation is not mechanically valid in every return series simply by multiplying a Sharpe ratio by T. The assumptions behind time aggregation matter, particularly serial correlation.

CORRECT INFORMATION

The standard T scaling is appropriate under restrictive assumptions such as independent, identically distributed returns. More generally, return aggregation and volatility estimation need to be treated separately and the time-series properties checked.

SOURCE / VERIFICATION

Andrew W. Lo, “The Statistics of Sharpe Ratios,” Financial Analysts Journal, 2002. CFA Institute — https://rpc.cfainstitute.org/research/financial-analysts-journal/2002/the-statistics-of-sharpe-ratios

RISKLENS NOTE

For the UI, label the period and annualisation convention rather than showing an apparently universal conversion.

5. Correlation heat-map palette · A2.2

Assessor-facing record of one deliberate error in the Zetheta training material.

INCORRECT STATEMENT

“Use a categorical palette for correlation heat maps.”

WHY IT IS WRONG

Correlation is not nominal category data. It has a meaningful midpoint at zero and direction from negative to positive.

CORRECT INFORMATION

Use a diverging color scheme when zero is a meaningful center. A diverging palette emphasizes the midpoint and the two directions away from it.

SOURCE / VERIFICATION

ColorBrewer, “Color scheme types” — https://colorbrewer2.org/learnmore/schemes_full.html

RISKLENS NOTE

The center value and direction should also remain understandable without color alone.

6. Universal 14 px text minimum · A2.5

Assessor-facing record of one deliberate error in the Zetheta training material.

INCORRECT STATEMENT

“14px minimum applies to ALL text including footnotes.”

WHY IT IS WRONG

WCAG 2.1 AA does not define a universal minimum font size in pixels. The assignment's appendix correctly treats 14 px as a recommendation rather than a universal WCAG requirement; the more precise WCAG requirements concern contrast and text resizing.

CORRECT INFORMATION

Do not describe 14 px as a universal WCAG minimum. WCAG 2.1 requires, among other things, sufficient text contrast and the ability to resize text to 200% without loss of content or functionality. Small captions may be appropriate when they remain readable and meet the relevant criteria.

SOURCE / VERIFICATION

W3C WCAG 2.1, SC 1.4.3 Contrast (Minimum) and SC 1.4.4 Resize Text — https://www.w3.org/TR/WCAG21/

RISKLENS NOTE

Zetheta's appendix notes 12 px as a practical body-text baseline. That is a design recommendation, not a WCAG pixel rule.

7. 6 px 'industry standard' grid · A3.4

Assessor-facing record of one deliberate error in the Zetheta training material.

INCORRECT STATEMENT

“The component library must use a 6-pixel base grid system.”

WHY IT IS WRONG

Calling 6 px the industry standard is incorrect. Common platform guidance uses 8-unit spacing as a primary rhythm; smaller increments can be used for fine alignment.

CORRECT INFORMATION

Use an 8-unit base rhythm for the RiskLens design system. Android explicitly documents an 8 dp grid for layout, components and spacing, with a 4 dp grid useful for smaller elements.

SOURCE / VERIFICATION

Android Developers, “Grids and units” — https://developer.android.com/design/ui/mobile/guides/layout-and-content/grids-and-units

RISKLENS NOTE

RiskLens therefore uses 8 px as the primary spacing rhythm, with 4 px available for micro-spacing.

8. Scaling the desktop UI for mobile · A4.3

Assessor-facing record of one deliberate error in the Zetheta training material.

INCORRECT STATEMENT

“Use CSS transform: scale() for mobile adaptation.”

WHY IT IS WRONG

Scaling an entire desktop canvas down preserves the desktop information architecture instead of adapting it to the constraints of a mobile context. It can also make controls and text harder to use.

CORRECT INFORMATION

Adapt the content, navigation and page flow for the mobile context. The W3C mobile guidance explicitly notes that the same information does not need to have exactly the same representation across devices and recommends adapting the experience to the delivery context.

SOURCE / VERIFICATION

W3C, Mobile Web Best Practices 1.0 and Mobile Web Application Best Practices — https://www.w3.org/TR/mobile-bp/ and https://www.w3.org/TR/mwabp/

RISKLENS NOTE

This is why RiskLens mobile screens are recomposed rather than scaled desktop frames.

9. 4.5:1 between adjacent chart series · A5.3

Assessor-facing record of one deliberate error in the Zetheta training material.

INCORRECT STATEMENT

“Every pair of adjacent bars must have a 4.5:1 contrast ratio…”

WHY IT IS WRONG

The statement applies the text contrast threshold to adjacent non-text graphics. WCAG 2.1 SC 1.4.11 sets a 3:1 minimum for meaningful non-text UI components and graphical objects against adjacent colors.

CORRECT INFORMATION

For meaningful non-text graphics, the relevant WCAG 2.1 AA minimum is 3:1 against adjacent colors where the contrast is required. The 4.5:1 threshold belongs to normal text under SC 1.4.3.

SOURCE / VERIFICATION

W3C WAI, “Understanding Success Criterion 1.4.11: Non-text Contrast” and WCAG 2.1 — https://www.w3.org/WAI/WCAG21/Understanding/non-text-contrast.html

RISKLENS NOTE

Use color, labels, patterns or other channels together when series must remain distinguishable.

10. Client-side cookies as secure authorization · A6.2

Assessor-facing record of one deliberate error in the Zetheta training material.

INCORRECT STATEMENT

“Client-side JavaScript … reads this cookie … This approach is both secure and performant because cookies are encrypted by the browser.”

WHY IT IS WRONG

A browser cookie or frontend JavaScript check cannot be trusted as the security boundary for authorization. Users can modify client-side state, so the server must enforce authorization independently.

CORRECT INFORMATION

Authorization decisions must be enforced server-side. Client-side role state can improve presentation and navigation, but it must never be the decisive control for protected resources.

SOURCE / VERIFICATION

OWASP Authorization Cheat Sheet — https://cheatsheetseries.owasp.org/cheatsheets/Authorization_Cheat_Sheet.html

RISKLENS NOTE

For RiskLens, role-based visibility in the UI should therefore be paired with backend permission checks.

11. Making every alert P1 Critical · A7.4

Assessor-facing record of one deliberate error in the Zetheta training material.

INCORRECT STATEMENT

“Set all alerts to P1 (Critical) priority by default…”

WHY IT IS WRONG

Making every alert critical defeats the purpose of severity classification and increases alert fatigue. Users become less able to distinguish genuinely consequential events from lower-priority information.

CORRECT INFORMATION

Use meaningful severity tiers, make only the most severe alerts interruptive, and reduce low-value alerts. The goal is to improve specificity and prioritisation, not to make everything urgent.

SOURCE / VERIFICATION

AHRQ PSNet, “Alert Fatigue” — https://psnet.ahrq.gov/primer/alert-fatigue ; PubMed review on alert-fatigue strategies.

RISKLENS NOTE

RiskLens should keep Critical, Warning and informational states distinct, with intervention intensity matching severity.

12. Leading usability tasks · A8.2

Assessor-facing record of one deliberate error in the Zetheta training material.

INCORRECT STATEMENT

“Tasks should be written in a leading manner that directs the participant to the correct answer.”

WHY IT IS WRONG

Leading tasks reveal whether people can follow instructions rather than whether the interface supports their natural decision-making. They also introduce bias.

CORRECT INFORMATION

Write realistic, scenario-based tasks that give participants enough context to act without telling them where to click or what control to choose. For example: “You are starting the morning risk check. Find the current VaR for the Global Macro portfolio.”

SOURCE / VERIFICATION

Nielsen Norman Group, “Turn User Goals into Task Scenarios for Usability Testing” — https://www.nngroup.com/articles/task-scenarios-usability-testing/ ; Digital.gov usability-test guidance — https://digital.gov/resources/how-conduct-usability-test

RISKLENS NOTE

The Zetheta brief itself asks for simulated scenarios; the task should describe the goal, not the intended interface path.

13. System Usability Scale (SUS) · A8.4

Assessor-facing record of one deliberate error in the Zetheta training material.

INCORRECT STATEMENT

“The SUS questionnaire consists of 15 questions on a 7-point Likert scale, with scores ranging from 0 to 150.”

WHY IT IS WRONG

That describes a different questionnaire format, not SUS. SUS is a compact standardized scale with a fixed item count and scoring method.

CORRECT INFORMATION

SUS has 10 items, each answered on a 5-point Likert scale. The final score ranges from 0 to 100. The scale was developed by John Brooke in 1986 while at Digital Equipment Corporation.

SOURCE / VERIFICATION

John Brooke, “SUS: A quick and dirty usability scale” (1986); supporting validation literature. — https://www.researchgate.net/publication/228593520_SUS_A_quick_and_dirty_usability_scale

RISKLENS NOTE

RiskLens usability reporting should therefore use the standard 10-item SUS scoring method.

14. Verification notes and source list

All 13 corrections are cross-checked against the Zetheta appendix. External sources are listed for independent verification.

#	Section	Verification source
1	A1.1	Nielsen Norman Group - 10 Usability Heuristics
2	A1.2	Fitts (1954), Journal of Experimental Psychology / DOI
3	A1.3	Acerbi & Tasche (2002); Inui & Kijima (2005)
4	A1.3	Lo (2002), Financial Analysts Journal / CFA Institute
5	A2.2	ColorBrewer - diverging scheme guidance
6	A2.5	W3C WCAG 2.1 - 1.4.3 and 1.4.4
7	A3.4	Android Developers - 8 dp grid
8	A4.3	W3C Mobile Web Best Practices
9	A5.3	W3C WCAG 2.1 - 1.4.11 Non-text Contrast
10	A6.2	OWASP Authorization Cheat Sheet
11	A7.4	AHRQ PSNet - Alert Fatigue
12	A8.2	Nielsen Norman Group + Digital.gov usability-task guidance
13	A8.4	John Brooke - SUS, 1986

Primary Zetheta source

Zetheta Project 1A training material, Appendix: “This appendix is for assessor reference. It lists all 13 deliberate errors and their corrections.”

Submission note: Keep this log with the final repository documentation so each correction can be traced back to the training material and independently verified.